

PROBLEMS, PROBLEM SPACES, & SEARCH IN AI

Introduction

- ❖ Artificial intelligence's initial goal is to build machines capable of carrying out tasks that usually call for human intelligence.
- ❖ Among the core functions of AI is real-life problem-solving.
- ❖ Understanding "problems," "problem spaces," and "search" is fundamental to comprehending how AI systems handle and resolve challenging jobs in the current situation.

Problems in AI

- ❖ A problem is a particular task or challenge that calls for decision-making or solution-finding.
- ❖ In AI, an issue is simply a task that needs to be completed; these tasks can be anything from straightforward math problems to intricate decision-making situations.
- ❖ AI encompasses various jobs and challenges, from basic math operations to sophisticated ones like picture recognition, natural language processing, gameplay, and optimization.
- ❖ Every problem has a goal state that must be attained, a defined set of initial states, and potential actions or moves.

Problems in AI

- ❖ **Important Components of Problems in AI:**
 - ❖ **Initial State:** The state of the issue as it first arises.
 - ❖ **Goal State:** The idealized final state that delineates a problem-solving strategy.
 - ❖ **Operators:** The collection of maneuvers or actions that can be used to change a state.
 - ❖ **Restrictions:** Guidelines or limitations must be adhered to solve the problem.
- ❖ **Example:** In a chess game, the pieces' beginning positions on the board represent the initial state, a checkmate is the objective state, the permissible moves made by the pieces represent the operators, and the chess rules represent the constraints.

Problem Spaces in AI

- ❖ **Problem Space**, is the set of all potential states, actions, and transitions that might arise when trying to solve a particular problem.
- ❖ It depicts the whole range of feasible fixes and routes from the starting point to the desired destination.
- ❖ **Problem Space**, can also be said to be an abstract representation of every conceivable state and all possible transitions between them for a particular problem.
- ❖ It is a conceptual landscape in which all points signify various system states, and all possible operations or activities are represented by the paths connecting the points.
- ❖ **Important Components of Problem Spaces in AI:**

Problem Spaces in AI

- ❖ **States:** Every scenario or configuration that could arise within the issue.
- ❖ **State Space:** The collection of all states that an operator sequence can apply to get from the starting state.
- ❖ **Paths:** Paths are sets of states that connect the starting state to the destination state through operators.
- ❖ In the case of route planning, for instance, the issue space consists of all potential locations on the map represented as states and all legitimate routes or paths connecting them as actions.
- ❖ Example, in a maze-solving problem, the problem space consists of the maze itself (state space), all potential positions within the maze (states), and the paths that travel from the start to the exit (paths) in the maze.

Search in AI

- ❖ **Search** is the practice of searching for a set of steps or movements that will get you to the desired outcome or a workable solution.
- ❖ Within AI, search algorithms are employed to methodically traverse the problem domain and identify routes or resolutions that fulfill the problem's limitations and goals.
- ❖ Search algorithms are used in AI to effectively explore issue domains.
- ❖ Search strategies can be generically categorized as **informed (heuristic)** and **uninformed (blind)**.

Problem Solving in AI

- ❖ To solve different types of AI problems, there are different types of AI techniques. These techniques use following steps:
 1. **Define the problem:** This definition must include precise specifications of what the initial situation(s) will be as well as what final situations constitute. By defining it properly, one can convert it into the real workable states.
 2. **Analyze the problem:** Identify the techniques to be used to solve the given problem. A few important features can have different techniques for solving the problem.
 3. Isolate and represent the task knowledge that is necessary to solve the problem.
 4. Choose the best problem-solving technique and apply on particular problem.

Problem Solving in AI

- ❖ The **reflex agents** are known as the simplest agents because they directly map states into actions. Unfortunately, they fail to operate in an environment where the mapping is too large to store and learn.
- ❖ **Goal-based agent**, on the other hand, considers future actions and the desired outcomes. Therefore, we will discuss one type of goal-based agent known as a **problem-solving agent**, which uses atomic representation with no internal states visible to the problem-solving algorithms.

Problem Solving in AI

- ❖ The problem-solving agent performs precisely by defining problems and its several solutions.
- ❖ According to psychology, *“a problem-solving refers to a state where we wish to reach to a definite goal from a present state or condition.”*
- ❖ According to computer science, a problem-solving is a part of AI which encompasses a number of techniques such as algorithms, heuristics to solve a problem.
- ❖ Therefore, a problem-solving agent is a goal-driven agent and focuses on satisfying the goal.

Problem Solving in AI

❖ Steps performed by Problem-solving agent

1. **Goal Formulation:** It is the first and simplest step in problem-solving. It organizes the steps/sequence required to formulate one goal out of multiple goals as well as actions to achieve that goal. Goal formulation is based on the current situation and the agent's performance measure.
2. **Problem Formulation:** It is the most important step of problem-solving which decides what actions should be taken to achieve the formulated goal. There are following five components involved in problem formulation:

Problem Solving in AI

- ❖ **Initial State:** It is the starting state or initial step of the agent towards its goal.
- ❖ **Actions:** It is the description of the possible actions available to the agent.
- ❖ **Transition Model:** It describes what each action does.
- ❖ **Goal Test:** It determines if the given state is a goal state.
- ❖ **Path cost:** It assigns a numeric cost to each path that follows the goal. The problem-solving agent selects a cost function, which reflects its performance measure.
Remember, an optimal solution has the lowest path cost among all the solutions.

Problem Solving in AI

- ❖ **Note:** Initial state, actions, and transition model together define the state-space of the problem implicitly.
- ❖ **State-space of a problem** is a set of all states which can be reached from the initial state followed by any sequence of actions.
- ❖ The state-space forms a directed map or graph where nodes are the states, links between the nodes are actions, and the path is a sequence of states connected by the sequence of actions.
- 3. **Search:** It identifies all the best possible sequence of actions to reach the goal state from the current state. It takes a problem as an input and returns solution as its output.

Problem Solving in AI

4. **Solution:** It finds the best algorithm out of various algorithms, which may be proven as the best optimal solution.
5. **Execution:** It executes the best optimal solution from the searching algorithms to reach the goal state from the current state.

❖ Example Problems

- ❖ There are two types of problem approaches:
 - ❖ **Toy Problem:** It is a concise and exact description of the problem which is used by the researchers to compare the performance of algorithms.

Problem Solving in AI

- ❖ **Real-world Problem:** Real-world based problems which require solutions. Unlike a toy problem, it does not depend on descriptions, but we can have a general formulation of the problem.
- ❖ **Some Toy Problems**
- ❖ **8 Puzzle Problem:** Here, we have a 3×3 matrix with movable tiles numbered from 1 to 8 with a blank space. The tile adjacent to the blank space can slide into that space. The objective is to reach a specified goal state like the goal state, as shown in the below figure.
- ❖ In the figure, the task is to convert the current state (Start) into goal state by sliding digits into the blank space.

Problem Solving in AI

7	2	4
5		6
8	3	1

Start State

	1	2
3	4	5
6	7	8

Goal State

- ❖ The problem formulation is as follows:
 - ❖ **States:** It describes the location of each numbered tiles and the blank tile.
 - ❖ **Initial State:** We can start from any state as the initial state.

Problem Solving in AI

- ❖ **Actions:** Here, actions of the blank space is defined, i.e., either left, right, up or down
- ❖ **Transition Model:** It returns the resulting state as per the given state and actions.
- ❖ **Goal test:** It identifies whether we have reached the correct goal-state.
- ❖ **Path cost:** The path cost is the number of steps in the path where the cost of each step is 1.
- ❖ **Note:** The 8-puzzle problem is a type of sliding-block problem which is used for testing new search algorithms in artificial intelligence.

Problem Solving in AI

❖ Water Jug Problem.

❖ **Statement:-** We are given 2 jugs, a 4 liter one and a 3-liter one. Neither has any measuring markers on it. There is a pump that can be used to fill the jugs with water. We can pour water out of a jug to the ground. We can pour water from one jug to another. How can we get exactly 2 liters of water into the 4-liter jugs?

❖ **Solution:-** The state space for this problem can be defined as:

❖ x -represents the number of liters of water in the 4-liter jug

Problem Solving in AI

- ❖ y - represents the number of liters of water in the 3-liter jug
- ❖ Therefore, $x = 0, 1, 2, 3$ or 4 and $y = 0, 1, 2$ or 3
- ❖ The initial state is $(0, 0)$. The goal state is to get $(2, n)$ for any value of ' n '.
- ❖ The various Production Rules that are available to solve this problem may be stated as given in the following table.

Problem Solving in AI

Rule No	Production Rule	Action
1	$(x,y) \text{ if } x < 4 \Rightarrow (4,y)$	Fill the 4 Liter jug
2	$(x,y) \text{ if } y < 3 \Rightarrow (x,3)$	Fill the 3 Liter jug
3	$(x,y) \text{ if } x > 0 \Rightarrow (0,y)$	Empty 4 Liter jug on ground
4	$(x,y) \text{ if } y > 0 \Rightarrow (x,0)$	Empty 3 Liter jug on ground
5	$(x,y) \text{ if } x+y \leq 4 \Rightarrow (x+y,0)$	Pour all the water from 3 liter jug into 4 liter jug
6	$(x,y) \text{ if } x+y \leq 3 \Rightarrow (0,x+y)$	Pour all the water from 4 liter jug into 3 liter jug
7	$(x,y) \text{ if } x+y \geq 4 \Rightarrow (4,y-(4-x))$	Pour water from 3 liter jug to 4 liter jug until 4 liter jug is full
8	$(x,y) \text{ if } x+y \geq 3 \Rightarrow (x-(3-y),3)$	Pour water from 4 liter jug to 3 liter jug until 3 liter jug is full
9	$(0,2) \Rightarrow (2,0)$	Pour 2 liter from 3 liter jug to 4 liter jug

Problem Solving in AI

❖ Solution 1.

Liter in 4 Liter Jug	Liter in 3 Liter Jug	Rule Applied
0	0	
0	3	2
3	0	5
3	3	2
4	2	7
0	2	3
2	0	5

Problem Solving in AI

❖ Solution 2.

Liter in 4 Liter Jug	Liter in 3 Liter Jug	Rule Applied
0	0	
4	0	1
1	3	8
1	0	4
0	1	6
4	1	1
2	3	8

Problem Solving in AI

❖ Solution 3.

Liter in 4 Liter Jug	Liter in 3 Liter Jug	Rule Applied
0	0	
4	0	1
1	3	8
0	3	3
3	0	5
3	3	2
4	2	7
0	2	3
2	0	5

Problem Solving in AI

- ❖ Thus, to solve any difficult, unstructured problems, we must provide a formal description to solve a problem and for that we must do following:
 1. Specify one or more states within that space that describes possible situations from which the problem-solving process may start, called as initial state.
 2. Specify one or more states that would be acceptable as solution to the problem (goal states).
 3. Define a state space that contains all the possible configuration of the relevant objects.
 4. Specify a set of rules that describes the action available thus, the problem can be solved by using the rules and with appropriate control strategy.

Problem Solving in AI

- ❖ For solving different kinds of problem, an agent makes use of different strategies to reach the goal by searching the best possible algorithms. This process of searching is known as **search strategy**.

Properties of Search Algorithms

- ❖ There are four important properties of search algorithms that are used to measure their performance:
 1. **Completeness:** It measures if the algorithm guarantees to find a solution (if any solution exist). A search algorithm is said to be complete if it guarantees to yield a solution for any random input if at least one solution exists.

Problem Solving in AI

2. **Optimality:** A solution discovered for an algorithm is considered optimal if it is assumed to be the best solution (lowest path cost) among all other solutions.
3. **Time Complexity:** The time taken by the algorithm to find a solution.
4. **Space Complexity:** The maximum storage space required during the search, as determined by the problem's complexity.

❖ The complexity of an algorithm depends on **branching factor** or **maximum number of successors**, **depth of the shallowest goal node** (i.e., number of steps from root to the path) and **the maximum length of any path in a state space**.

Problem Solving in AI

❖ Importance of Search Algorithms in Artificial Intelligence

1. **Solving problems:** Using logical search mechanisms, including problem description, actions, and search space, search algorithms in artificial intelligence improve problem-solving. Applications for route planning, like Google Maps, are one real-world illustration of how search algorithms in AI are utilized to solve problems. These programs employ search algorithms to determine the quickest or shortest path between two locations.
2. **Search programming:** Many AI activities can be coded in terms of searching, which improves the formulation of a given problem's solution.

Problem Solving in AI

3. **Goal-based agents:** Goal-based agents' efficiency is improved through search algorithms in artificial intelligence. These agents look for the most optimal course of action that can offer the finest resolution to an issue to solve it.
4. **Support production systems:** Search algorithms in artificial intelligence help production systems run. These systems help AI applications by using rules and methods for putting them into practice. Production systems use search algorithms in artificial intelligence to find the rules that can lead to the required action.

Problem Solving in AI

5. **Neural network systems:** The neural network systems also use these algorithms. These computing systems comprise a hidden layer, an input layer, an output layer, and coupled nodes. Neural networks are used to execute many tasks in artificial intelligence. For example, the search for connection weights that will result in the required input-output mapping is improved by search algorithms in AI.